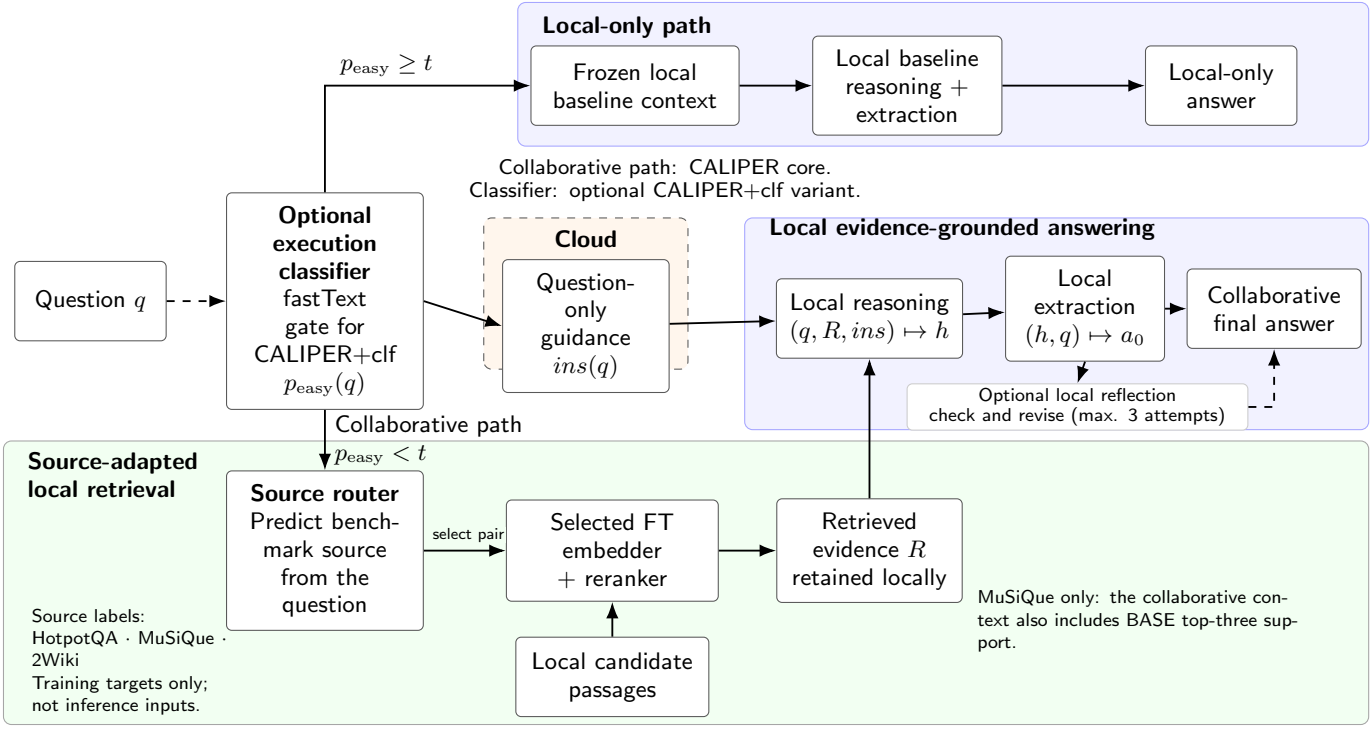

Local-only path
$p_{\text{easy}} \geq t$
Frozen local baseline context
Local baseline reasoning + extraction
Local-only answer
Collaborative path: CALIPER core.
Classifier: optional CALIPER+clf variant.
Question q
Optional execution classifier
fastText gate for CALIPER+clf
$p_{\text{easy}}(q)$
Cloud
Question-only guidance
$ins(q)$
Local evidence-grounded answering
Local reasoning
$(q, R, ins) \mapsto h$
Local extraction
$(h, q) \mapsto a_0$
Collaborative final answer
Optional local reflection
check and revise (max. 3 attempts)
Collaborative path
$p_{\text{easy}} < t$
Source-adapted local retrieval
Source router
Predict benchmark source from the question
select pair
Selected FT embedder + reranker
Retrieved evidence R retained locally
Local candidate passages
Source labels:
HotpotQA · MuSiQue · 2Wiki
Training targets only;
not inference inputs.
MuSiQue only: the collaborative context also includes BASE top-three support.